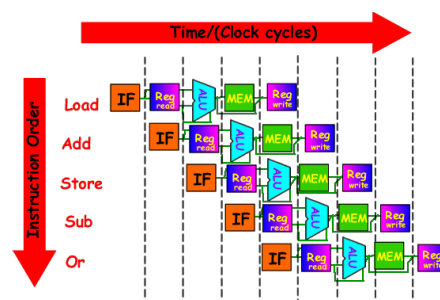


Week 2-4 Pipelining Overview.



Description

This module introduces the basic concepts of CPU Pipeline, increasing CPU performance by improving instruction throughput. We will start by reviewing the single-cycle RISC-V data and control path; to then talk about how to make it a pipelined CPU. The module will describe and analyze pipeline hazards: Structural, Data, and control; and will discuss solutions to them. This unit is three weeks in length, and you should expect to spend 8-12 hours per week completing the readings and activities.

Learning Outcomes

By the end of this week you will be able to:

1. Define the concept of a pipeline.
2. Compare Pipeline vs Single Cycle CPU
3. Develop a DataPath/ControlPath for RISC-V 32Bit ISA with 5 Stages Pipeline

The above Learning Objectives above align with the Course Learning Objective (list here):

1. To understand pipelining, its performance benefits, and associated design challenges.
2. To design and implement a simulator for a specific microprocessor architecture

This Week's Materials

The following materials will help you meet the learning objectives listed above

PowerPoint slides, which can be found here in pdf:

1. Slides (used in class) Pipeline Basics + Data Hazards and Control Hazards [3.0Pipeline&DataHazards.pdf](https://canvas.calpoly.edu/courses/186931/files/21847972/download?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847972/download?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847972/download?download_frd=1) (https://canvas.calpoly.edu/courses/186931/files/21847972/download?download_frd=1)
2. OtterPipeline Extra Starting Code Template [link](https://docs.google.com/presentation/d/1H_S6W6Q0OBnOeRPGz3C8l5Z1-lm7lyDBpwAemaxVC6U/edit#slide=id.g32bd6927f4b_0_13) [↗](https://docs.google.com/presentation/d/1H_S6W6Q0OBnOeRPGz3C8l5Z1-lm7lyDBpwAemaxVC6U/edit#slide=id.g32bd6927f4b_0_13) (https://docs.google.com/presentation/d/1H_S6W6Q0OBnOeRPGz3C8l5Z1-lm7lyDBpwAemaxVC6U/edit#slide=id.g32bd6927f4b_0_13)
3. RISC V Simulators on web educational
 1. [WebRISC-V - RISC-V PIPELINED DATAPATH SIMULATION ONLINE](https://web riscv.altervista.org/) [↗](https://web riscv.altervista.org/) (<https://web riscv.altervista.org/>)
 2. [SonicRV](https://sonic-rv.ics.jku.at/example/ics-nex-rv32i-pl/all_instructions) [↗](https://sonic-rv.ics.jku.at/example/ics-nex-rv32i-pl/all_instructions) (https://sonic-rv.ics.jku.at/example/ics-nex-rv32i-pl/all_instructions)
4. Book chapter on Pipeline DataPath and Control Path Implementation [ImplementationsDetails_Ch7.pdf](https://canvas.calpoly.edu/courses/186931/files/21847957/download?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847957/download?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847957/download?download_frd=1) (https://canvas.calpoly.edu/courses/186931/files/21847957/download?download_frd=1)
5. Book chapter Pipeline Verilog Implementation Details (Includes review for RISC-V data path from 233) [verilogDetails.pdf](https://canvas.calpoly.edu/courses/186931/files/21847971/download?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847971/download?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847971/download?download_frd=1) (https://canvas.calpoly.edu/courses/186931/files/21847971/download?download_frd=1)
6. Loop Unrolling Notes [loop_unrolling_notes_7-13-2020.pdf](https://canvas.calpoly.edu/courses/186931/files/21847971/download?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847971/download?wrap=1>)

[files/21847958?wrap=1](https://canvas.calpoly.edu/courses/186931/files/21847958/download?download_frd=1)) [↓](https://canvas.calpoly.edu/courses/186931/files/21847958/download?download_frd=1) (https://canvas.calpoly.edu/courses/186931/files/21847958/download?download_frd=1)

2. Links to reading resources here:

1. Fantastic review of RISC_V single cycle, multicycle, and pipeline implementations with the Verilog code in section 7.6; it should be a good read for all of us and especially for the students that took this class "long ago" and need to read/refresh. We will cover in class chapters 7.5 (pipeline), 7.6(Verilog code) [7---](#) [Microarchitecture 2022 Digital-Design-and-Computer-Architecture.pdf](https://canvas.calpoly.edu/courses/186931/files/21847966?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847966?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847966/download?download_frd=1) (https://canvas.calpoly.edu/courses/186931/files/21847966/download?download_frd=1)
2. [Basic Pipeline Concept COD ch 4.5.pdf](https://canvas.calpoly.edu/courses/186931/files/21847961/download?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847961/download?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847961/download?download_frd=1) (https://canvas.calpoly.edu/courses/186931/files/21847961/download?download_frd=1)
3. [Data Hazards COD 4.7.pdf](https://canvas.calpoly.edu/courses/186931/files/21847823?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847823?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847823/download?download_frd=1) (https://canvas.calpoly.edu/courses/186931/files/21847823/download?download_frd=1)
4. [Control Hazards COD 4.8.pdf](https://canvas.calpoly.edu/courses/186931/files/21847969?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847969?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847969/download?download_frd=1) (https://canvas.calpoly.edu/courses/186931/files/21847969/download?download_frd=1)
5. Optimal Pipeline Depth [hartsteina optimum pipeline.pdf](https://canvas.calpoly.edu/courses/186931/files/21847964?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847964?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847964/download?download_frd=1) (https://canvas.calpoly.edu/courses/186931/files/21847964/download?download_frd=1)
3. (<https://canvas.calpoly.edu/courses/186931/files/21847961/download?wrap=1>) Practice exercises: [Pipeline Hw practice](https://canvas.calpoly.edu/courses/186931/pages/pipeline-hw-practice?wrap=1) (<https://canvas.calpoly.edu/courses/186931/pages/pipeline-hw-practice?wrap=1>)
4. Go to the following webpage: [Pipeline Visual Simulation](https://vhosts.eecs.umich.edu/370simulators/pipeline/simulator.html#) (<https://vhosts.eecs.umich.edu/370simulators/pipeline/simulator.html#>) It is a pipeline visual simulator where you can change/design your own parameters for the instruction and visualize how the instruction gets executed through the DataPath.

Activities and Formative Assessment

The following activities will help you master the learning objectives listed above:

1. (Student interaction learning activity):

After attending the class and reviewing the week's materials (webpage, pdf file, and book chapter) submit a written reflection of what you have learned about pipelines. Include the following: How would you describe a pipeline to a friend or peer? Why is pipelining fundamental for CPU execution time improvement? " Discussion: Basic Pipeline questions" Create Discussion [Week 2 Discussion | Pipeline](https://canvas.calpoly.edu/courses/186931/discussion_topics/898831?wrap=1) (https://canvas.calpoly.edu/courses/186931/discussion_topics/898831?wrap=1)

2. (Reflective Analysis) :

- Discussion W3 [Discussion Pipeline Basics week 3](https://canvas.calpoly.edu/courses/186931/discussion_topics/898830?wrap=1) (https://canvas.calpoly.edu/courses/186931/discussion_topics/898830?wrap=1)

3. (Formative assessment):

- Complete the Quiz:
 - Please complete the **Quiz: Computer Architecture Basics**. [Quiz 1](https://canvas.calpoly.edu/courses/186931/assignments/1612092?wrap=1) (<https://canvas.calpoly.edu/courses/186931/assignments/1612092?wrap=1>)
 - Pipeline Basics and Data Hazards: [Quiz Pipeline Basic Problems Week 3](https://canvas.calpoly.edu/courses/186931/assignments/1612096?wrap=1) (<https://canvas.calpoly.edu/courses/186931/assignments/1612096?wrap=1>)
 - Pipeline Hazards and Implementation Details: [Quiz Pipeline Problem Week 4](https://canvas.calpoly.edu/courses/186931/assignments/1612093?wrap=1) (<https://canvas.calpoly.edu/courses/186931/assignments/1612093?wrap=1>)
- Pipeline Assignment No Hazards [Lab 3: 5-stage Basic Pipeline](https://canvas.calpoly.edu/courses/186931/assignments/1612087?wrap=1) (<https://canvas.calpoly.edu/courses/186931/assignments/1612087?wrap=1>)
- Pipeline Assignment WITH hazards detection and forwarding [Lab 4: Pipeline w/ forwarding, hazard handling](#)

<https://canvas.calpoly.edu/courses/186931/assignments/1612088?wrap=1>